Name: ____________________ **Date:** ____________________

Collaborators: ____________________ **Lab Section:** ____________________

Postlab Exercises (20 points)

Submit the postlab to the TA at the end of the lab.

Data Acquisition - AUTO Mode (10 points)

Question 4 **7 points**

Using the vertical cursor on the oscilloscope, measure the voltage spacing between adjacent dips in the trace.

Location of first dip = ____________ V

Data table 1: *Voltage period measurements from the Franck-Hertz Kit.*

Pairs of dips	$\Delta V = V_{i+1} - V_i$ [V]	Average ΔV [V]
1-2		
2-3		
3-4		
4-5		
5-6		
6-7		
7-8		

Question 5 **1 points**

(a) ($1/2$ point) Calculate the standard deviation in average voltage spacing.

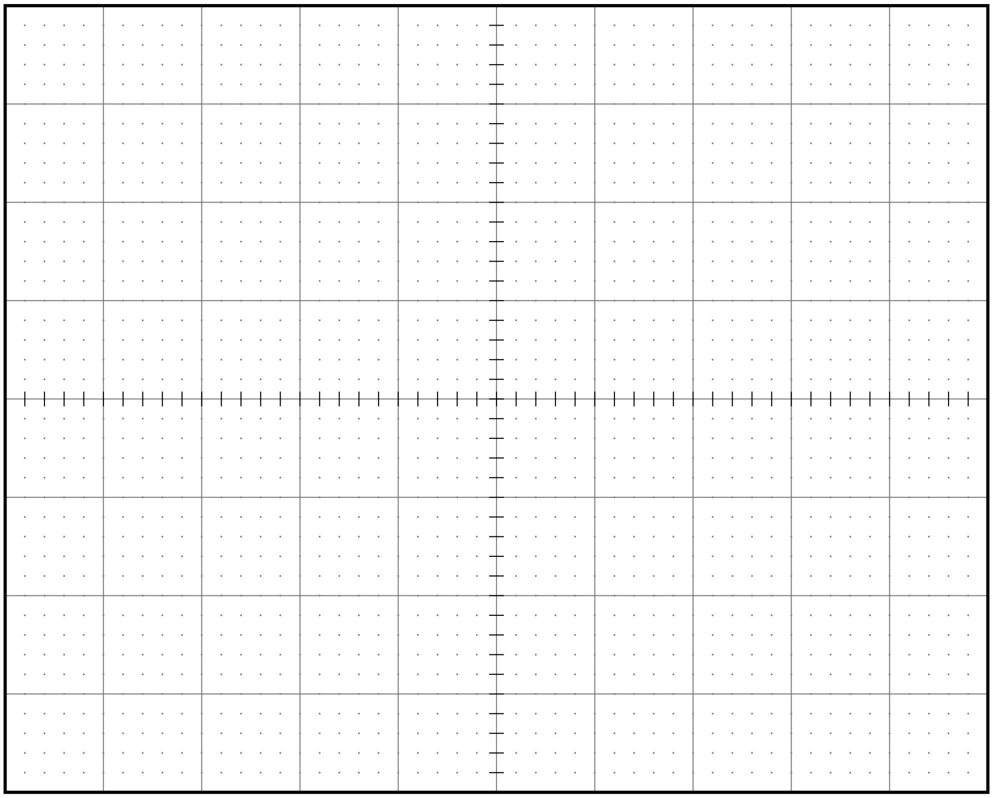
(b) ($1/2$ point) What does average voltage spacing physically correspond to?

Question 6 **2 points**

Sketch the current versus grid voltage as seen in the oscilloscope. Add a title, axis labels, and units.

Alternatively, you can save the oscilloscope window in picture format and attach it to your hand-in.

Graph 1: *Current versus grid voltage* $V_{G2,K}$.



Data Analysis (10 points)

Question 7 **1 point**

How much energy is required to excite an electron in the valence shell of Argon to its first excited state?

Question 8 **2 points**

Atoms do not stay in their excited states for long. After some time, the excited Ar atoms return to the ground state by releasing excess energy in the form of electromagnetic radiation. If you were to measure the frequency of radiation from your Franck-Hertz experiment, what would be the frequency?

Question 9 **2 points**

Suppose you want to excite Ar to the first excited state by shining electromagnetic radiation on it. What should be the wavelength of the radiation?

Question 10 **1 point**

If everything went correctly, you should have seen that the current at the higher-order dips was greater compared to the current at the lower-order dips. What could be the reason?

Question 11 **1 point**

How might contamination of other gases in the tube affect your result?

Question 12 **3 points**

What is your overall conclusion about energy levels in the atom/atomic model from this experiment? Describe its connection to Bohr's atomic model.